

DESIGNING & MANUFACTURING IN THE UK SINCE 1923



P852K

Next Generation
LED Luminaire



This luminaire complies with ETL guidelines for White Light Emitting Diode Lighting Units and is eligible for the Enhanced Capital Allowance (ECA) scheme.



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INTRO



P852K is a high quality, low cost solution to replace existing sodium discharge and compact fluorescent luminaires on minor roads and residential areas.

P852K has been designed to meet most demanding lighting requirements, being easy to install and maintain. It combines latest LED light source with state-of-the-art design, achieving long life for both LEDs and the drivers. The installation is simple and fast, and the luminaire is easily upgradable on-site if required.

LUMINAIRE LUMINOUS FLUX	up to 4250 lm
LUMINAIRE EFFICACY	up to 115 lm/W
LUMEN MAINTENANCE	L90 > 100,000 hrs at full power, Ta = 25°C
PHOTOMETRIC OPTIONS	6 distributions

KEY BENEFITS

- Elegant, state-of-the-art design
- Replace conventional LPS/HPS/CFL lanterns at low cost
- Future-proof and upgradable on site
- Superior luminaire efficacy up to 115 lm/W
- Wide range of optics and lumen packages
- Advanced thermal management
- Maximised savings on energy and maintenance costs
- Contractor-friendly installation and maintenance
- Minimal total cost of ownership
- Up to S3/P3 lighting class applications
- Dark sky friendly and no upward light
- Flexible and intelligent lighting control options
- IP66 ingress protection
- 100% recyclable

IMPROVED SERVICEABILITY



- Tool-less access
- Easy, fast wiring and installation
- Contractor-friendly maintenance
- Quick replacement for LED and Driver compartment
- Automatic electrical isolation when opened
- Easy electrical testing without altering wiring

FLEXIBLE MOUNTING OPTIONS

Universal SE/PT spigot caps to suit 34-42mm, 42-60mm and 60-76mm nominal diameter spigots providing -10°, -5°, 0°, +5° and +10° tilt in both post top and side entry arrangements with permanent indication on the luminaire.

Ø 60 - 76MM X 76MM POST-TOP



Ø 34 - 42MM X 100MM SIDE-ENTRY / POST-TOP



Ø 42 - 60MM X 100MM SIDE-ENTRY / POST-TOP



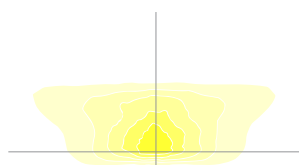
EXCEPTIONAL OPTICAL PERFORMANCE



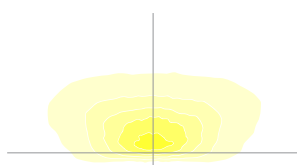
Standard Neutral White LEDs (CCT = 4000K)
Optional Warm White LEDs (CCT = 3000K)
Colour Rendering Index > 70
Improved mesopic vision
High quality PMMA Lenses
Exceptional Uniformity
Dark Sky Friendly (no upward light)

OPTICAL DISTRIBUTIONS

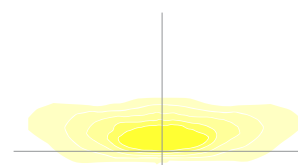
P852K offers a wider choice of optics and lumen packages. High efficacy optics six different types of distributions allow even the most challenging of schemes to be lit with maximum energy efficiency.



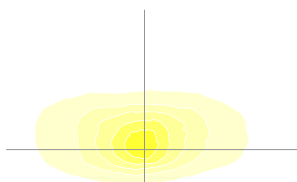
R1



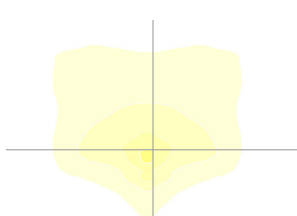
R2B



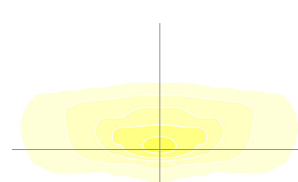
R3B



C1

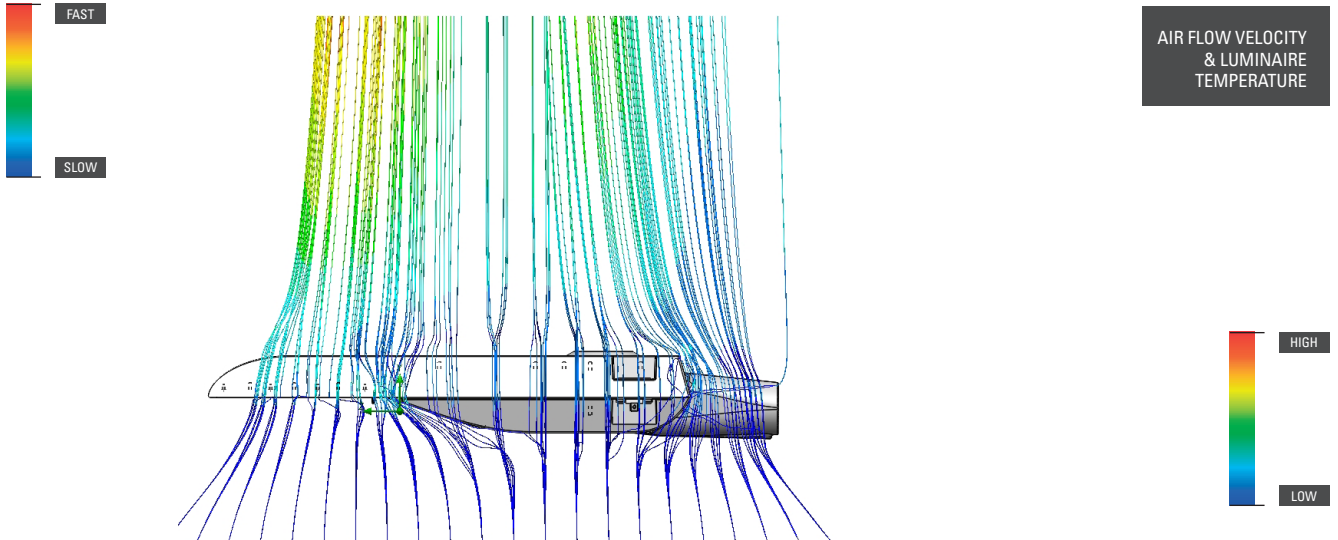


F2



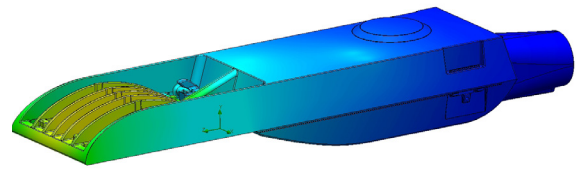
P4

ADVANCED THERMAL MANAGEMENT



P852K's simple design quickly transfers heat from the LEDs to the walls of the vertical vents, which dissipates the heat into the air and away from the luminaire.

The complete separation of the driver compartment from the LEDs keeps the driver very cool, significantly increasing the luminaire operating life even in high ambient operating temperatures.



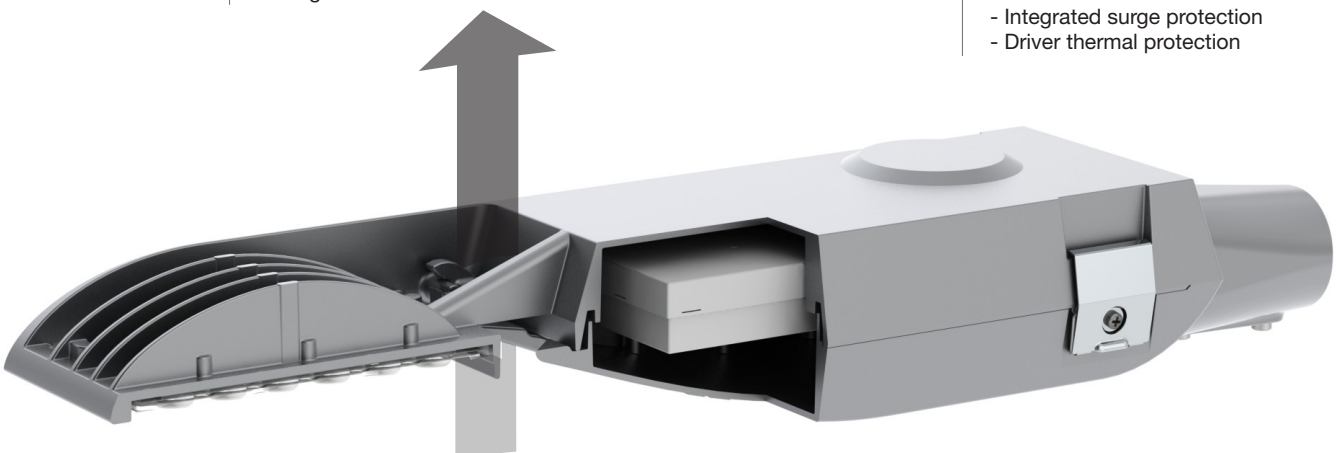
LUMINAIRE TEMPERATURE RESULTS FROM CFD

AIR VOID

- Both gear and optical compartments are separated in order to optimise thermal management

PROGRAMMABLE DRIVER

- Single level or multi-level dimming
- Adjustable Output Current (AOC)
- Constant Light Output (CLO)
- DALI dimmable
- Integrated surge protection
- Driver thermal protection



PMMA LENSES

- Increased luminaire efficacy
- Dark sky friendly
- Easy cleaning externally

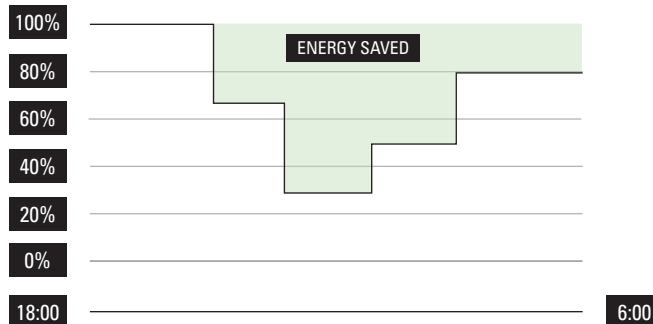
LUXEON® TX LEDs

- Superior light output
- High efficacy
- Proven reliability
- Tight CCT control

MULTI-STEP DIMMING

The programmable driver incorporates the multi-step dimming feature, a programmable 5-step dimming system which will generate substantial energy savings by providing the precise amount of light at the right time. The times and light levels are fully flexible to suit the required lighting profile.

The driver is able to calculate the virtual clock time by analysing the duration of operation of the driver from the previous 3 days and sets the times of 5 light level steps accordingly.

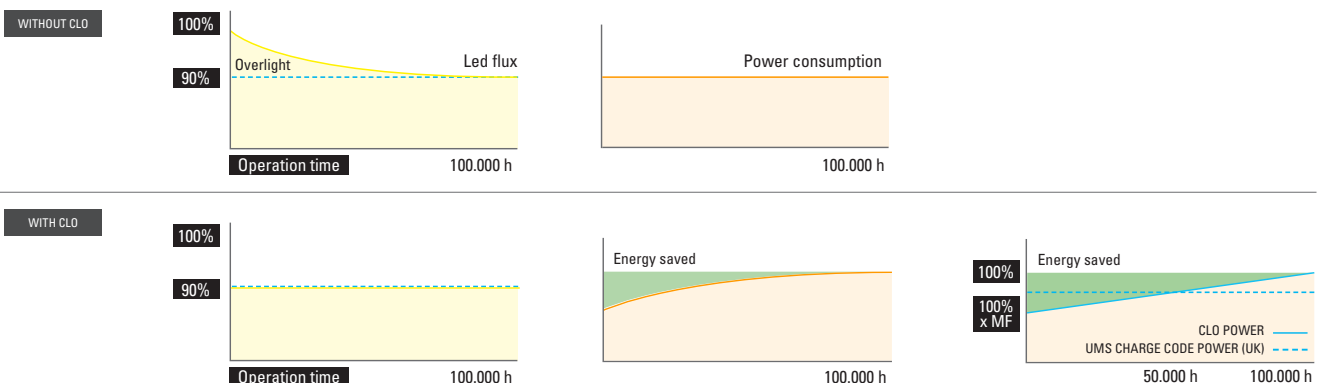


CONSTANT LIGHT OUTPUT (CLO)

All light sources experience lumen depreciation - a reduction in light output over time, which means the system would consume more power than necessary to meet the required light levels at the end of the lamp's useful life (e.g. L90).

The drivers of the P852K can be programmed to ensure that the LEDs will always deliver the necessary light level, by increasing the operating current over time to compensate for the LED lumen depreciation.

Over-lighting at the beginning is taken away and this feature can produce extra energy saving and extend the lifetime of the system.



PROGRAMMABLE LIGHTING CONTROLS

The programmable driver enables CU Phosco® Lighting to adjust the light level to match a specific application with optimised energy savings. The various control options offer different levels of energy savings, from simple stand-alone controls to more advanced networked Central Management Systems (CMS).

P852K is currently compatible with the following CMS:

Harvard LeafNut
Mayflower
Philips Starsense
Telensa PLANet
Zodion Vizion
Charles Endirect CELnet
Telematics

CONTROL SYSTEM	BENEFITS	FUNCTIONALITY	RELATIVE SAVING	WITH CLO
Photocell	Standard control	Switch on/off with ambient light level	0%	up to 10%
Multi-step dimming	Substantial energy saving	Programmable dimming (up to 5 steps)	up to 20%	up to 30%
Wireless CMS	Full control and monitoring of each individual luminaire	DALI and 1-10V dimming inputs with full CMS functionality	up to 40%	up to 50%

SCOTOPIC / PHOTOPIC (S/P) RATIO TO BS 5489-1:2013

Recent scientific research shows a correlation between the spectral power distribution of a light source and the visual performance under low lighting levels associated with mesopic vision.

For the levels associated with lighting residential and minor roads to the S classes from BS EN 13201-2:2003 and P classes from

CIE 115:2010, the target illuminance for a class can be adjusted according to the S/P ratio.

The S/P ratio of P852K neutral white LEDs is 1.64. The target illuminance for the P classes are shown here. For more information, refer to ILP Professional Lighting Guide 03: Lighting for subsidiary roads.

LIGHTING CLASS	BENCHMARK RA < 60		S/P RATIO = 1.64 AND RA ≥ 60	
	E	E _{min}	E	E _{min}
P1	15.0	3.0	12.76	2.55
P2	10.0	2.0	8.06	1.61
P3	7.5	1.5	5.86	1.17
P4	5.0	1.0	3.66	0.73
P5	3.0	0.6	1.98	0.40
P6	2.0	0.4	1.20	0.40

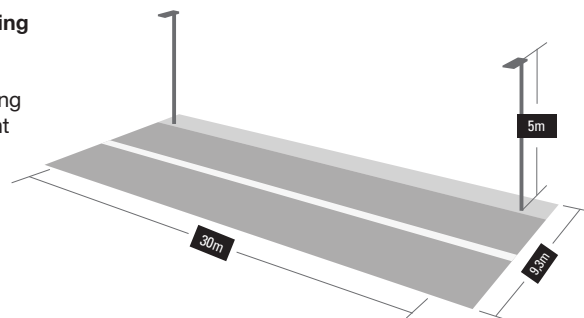
Modification of
BS 5489-1:2013
Table A.7

S AND P CLASS SCHEME EXAMPLE

P852K can replace a conventional 35W LPS (SOX) or 55W CFL luminaire with better performance. 77% energy savings are achievable depending on column spacing, road configuration and lighting class with the added comfort of white light. Further savings can be achieved using controls like Dynadimmer or a Central Management System.

Road refurbishment S4/P4 lighting class (EN13201/BS5489-1:2013)

Luminaire replacement with existing column at 30m spacing, 5m height and single sided arrangement.



	E (LX)	EMIN (LX)				
Target (S4)	5.0 - 7.5	1.0				
Target (P4)	3.66 - 5.49	0.73				
			W (SYSTEM)	LUMINAIRE EFFICACY	W / KM	ENERGY SAVINGS VS 35W LPS
35W LPS Luminaire	5.1	1.1	65*	49 lm/W	2167	-
35W LPS Luminaire	5.1	1.1	39**	82 lm/W	1300	40%
55W CFL Luminaire	5.2	1.1	62	54 lm/W	2067	5%
P852K***	3.75	0.84	16	109 lm/W	533	75%

*Using electromagnetic ballast

**Using HF electronic ballast

***P852K running at 400mA with CLO, Wide distribution and S/P ratio 1.64

TOTAL COST OF OWNERSHIP

While CFL technology has a low initial cost, it requires frequent maintenance, resulting in a high total cost of ownership.

P852K with dimming and CLO options delivers an attractive total cost of ownership package making it extremely competitive for invest-to-save replacement schemes.



* Based on the example above, TCO calculated over 20 years

P852K SPECIFICATION

Light source
Number of LEDs
Power Consumption
Luminaire Luminous Flux
Luminaire Efficacy
Driver Current (in 50mA steps)
Lumen Maintenance Output (Ta = 25°C)
Operating Temperature (for 100,000 hrs)
Maximum Operating Temperature
Weight (Total)

Lumileds LUXEON® TX LEDs
6
8 ~ 21W
730 ~ 2,150 lm
up to 103 lm/W
300mA ~ 1000mA
L95 > 100,000 hours
-30°C to +40°C (1000mA)
40°C
5kg

12
7 ~ 38W
1040 ~ 4,250 lm
up to 115 lm/W
200mA ~ 1000mA
L90 > 100,000 hours
-30°C to +40°C (850mA)
40°C
5kg

Correlated Colour Temperature

Neutral White, 4000K
Optional, Warm White, 3000K
G3
> 70
PMMA Lenses
I
DALI or Step-dimming

Glare Rating
Colour Rendering Index
Optical Cover
Electrical Class
Control System Input

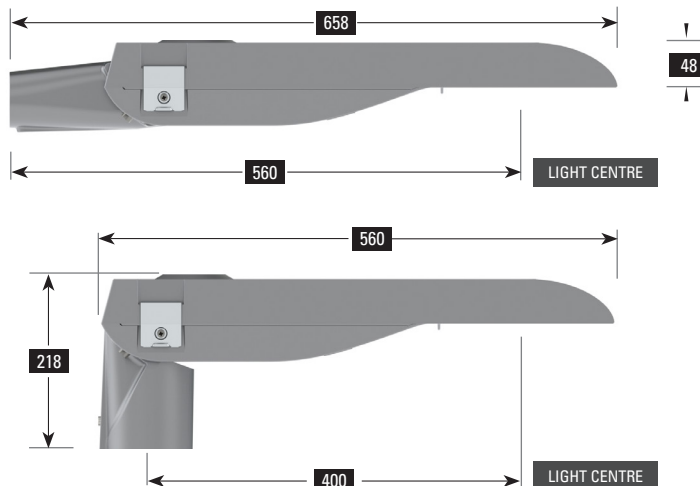
Surge Protection
Dimming Control
Lighting Regulation

8 kV Comm. Mode 6 kV Diff. Mode to IEC 61000-4-5; 10 kV Single Pulse
Multi-step dimming
Mini Photocell • NEMA Socket • Wireless CMS options

Installation Height
Installation

3 ~ 6m
Ø 34-42mm x 100mm Long SE • Ø 42-60mm x 100mm Long SE/PT
Ø 60-76mm x 76mm Long PT
-10°, -5°, 0°, 5°, 10°
High pressure die cast aluminium (Housing)
Polyester powder coat cured under heat
Light grey (RAL 7035), other RAL colours available on request
IP66
0.04m²

Post Top / Side Entry Tilt
Material
Finish
Colours
Ingress Protection
Wind Area (EPA)



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